AAE5303 Robust Control Technology in Low-Altitude Aerial Vehicle

Post-Lesson Reflection Report

Student Name: *CHEN Xiting*

Student ID: *25091409G*

Group Number: *X.AeroFusion*

Date: *2026.04.26*

Section 1: AI Usage Experience

As a complete beginner with no coding background， I have to admit that for nearly every step of finishing the project, from understanding the project content to running the outcome by myself, I relied quite heavily on Cursor AI assistance. During the 3D reconstruction phase (group assignment), even though I watched the tutorial video from the TA, I still didn't understand the process of finishing the project, such as why I needed to use Colmap and how to achieve a better 3D Gaussian model. So I just gave the document from my group member (VO phase) and asked AI to tell me the specific process of using this data to reconstruct a model and what other files I might need. AI just gave me a detailed explanation of the pipeline from input to output, as well as the purpose of each step. I quickly understood that I needed to use Colmap to obtain the sparse point cloud as the input for using Openspalt by feature extraction， sequential matching, and mapping process. That was exactly the SFM process I learnt in class!

Overall, AI helped me to systematically understand the project and enabled me to break down the task into specific executions, which I thought was crucial to achieve a good outcome. More importantly, AI just built a bridge to the theoretical knowledge and abstract concepts I learned in class and the implementation process. During the project, I had several moments of sudden realization about concepts I hadn't understood in class, which really deepened my understanding of the technology.

Section 2: Understanding AI Limitations

When I used the code from AI to finish the Colmap process and got the files of reconstructed camera pose and sparse point cloud, I directly gave the data to OpenSplat, and it turned out to be an incorrect camera model. So I checked the camera models with existing configurations and found it was Full_opencv, which also meant the images had distortion. And this is because AI incorrectly suggested me to keep the camera intrinsic configuration from the previous VO

process to improve reconstruction accuracy without checking whether VO used the original camera image (without distortion correction) or not. As OpenSplat only supports the Pinhole camera model, I used “colmap image_undistorter” to export the current model and got the distortion-corrected images, the corresponding new camera intrinsic (Pinhole), and the sparse model, which can be used by OpenSplat for 3D reconstruction.

Failed
Error Mode :Incorrect Camera Model

```
# Camera list with one line of data per camera:
#   CAMERA_ID, MODEL, WIDTH, HEIGHT, PARAMS[]
# Number of cameras: 1
1 FULL_OPENCV 1224 1024 722.21500000000003 722.
```

Unsupported model for OpenSplat

```
11    Camera1.k1: -0.0560
12    Camera1.k2: 0.1180
13    Camera1.p1: 0.00122
14    Camera1.p2: 0.00064
15    Camera1.k3: -0.0627
```

Mode with distortion

```
Undistorting image [1280/1288]
Undistorting image [1281/1288]
Undistorting image [1282/1288]
Undistorting image [1283/1288]
Undistorting image [1284/1288]
Undistorting image [1285/1288]
Undistorting image [1286/1288]
Undistorting image [1287/1288]
Undistorting image [1288/1288]
Writing reconstruction...
Writing configuration...
Writing scripts...
Elapsed time: 0.880 [minutes]
```

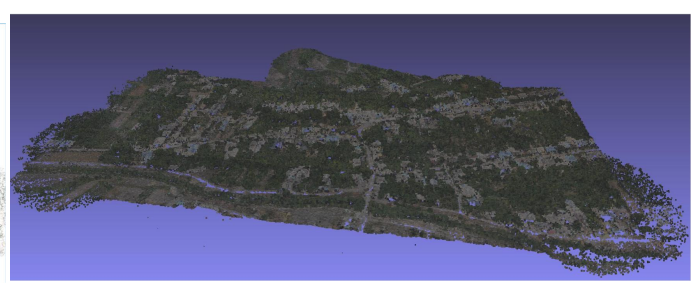
```
ccxx@ccxx:~/colmap_ws$ tmpdir=$(mktemp -d)
colmap model_converter \
  --input_path dense/sparse \
  --output_path "$tmpdir" \
  --output_type TXT
head -n 8 "$tmpdir/cameras.txt"
rm -rf "$tmpdir"
# Camera list with one line of data per camera:
#   CAMERA_ID, MODEL, WIDTH, HEIGHT, PARAMS[]
# Number of cameras: 1
1 PINHOLE 1221 1020 722.21500000000003 722.16999999999996
```

Distortion correction and Convert Camera model

Section 3: Engineering Validation

After I finished the image distortion correction and camera model conversion phase suggested by AI, I was not sure about the quality of my updated data, which actually set the foundation of the basic precision of the reconstruction model. I validated the Colmap output by: 1) checking the number of images to ensure no data loss. 2) Use Meshlab to visualize the sparse point cloud and check whether it has outlier 3D Points. 3) Compare the camera trajectory in the sparse model with the estimated trajectory from VO, as well as the ground truth trajectory.After verifying the output data of Colmap, I am certain that the accuracy of the 3D Gaussian model will not be affected by the original data.

Camera trajectory



Sparse point cloud

Section 4: Problem-Solving Process

I thought the most challenging problem for me to do the 3D reconstruction project was to optimize and adjust parameters for a better quality model. At the beginning, I even had no idea about what the specific criteria of a good outcome were, because we didn't have a quantity leaderboard. So I asked AI how to justify a good model and the strategy of adjusting the parameters for optimization, as there were more than a dozen parameters. Then, I knew the simplest way was to visualize the model to justify by eye: the clearer, the more realistic, the better.

And the optimization strategy was from the overall to the details. The first step is to quickly verify the availability of the dataset with a small iteration number (2000) and other default parameters. Then, I gradually increased the iteration number, trying to find a model with higher accuracy.

OpenSplat parameter:

- Num-iters: 2000,10000,20000,30000 (default 30000)
- Downscale-factor: 2 (default 1)
- Sh-degree: default 3
- Ssim-weight: default 0.2
- Save-every: default: -1
- Densify-grad-thresh: default: 0.0002
- Densify-size-thresh: default: 0.01
- Refine-every: default 100

First round optimization

However, the outcome was not the more iterations, the higher accuracy. I was quite confused and gave the screenshot to AI for analysis. AI speculates that the reason might be due to overfitting/the continuous increase of floating-point numbers in the later stages. AI suggested me to reduce the iteration number to reduce later-stage noise and also adjust other parameters to optimize the obvious blur in vegetation.

-n 22000 \	30000	
-d 2 \		
--densify-grad-thresh 0.0003 \	0.0002	
--refine-every 150 \	100	previous version
--warmup-length 800 \	500	
--reset-alpha-every 40 \	30	
--stop-screen-size-at 3000 \	4000	
--ssim-weight 0.15 \	0.2	
--save-every 2000 \		

Second round optimization

After I adjusted the parameters, I did get a better outcome this turn. But I still didn't completely understand the purpose of adjusting a specific parameter, so I asked AI to give me a precise explanation of each adjustment.

This configuration, compared to your original default (only changing -n 30000 -d 2), mainly modifies these aspects:

-n 22000 (reduced from 30000 to 22000)
Purpose: Reduces overfitting/the continued growth of floating points in later stages.
Reason: In your image, the quality improvement from 20k to 30k is limited, but noise increases.

--densify-grad-thresh 0.0003 (default 0.0002, increases the threshold)
Purpose: Adds Gaussian elements more "cautiously," reducing noise.
Reason: Lower thresholds make densification easier, potentially amplifying noise as well.

--refine-every 150 (default 100, reduces refine frequency)
Purpose: Reduces the number of split/copy/pruning triggers, resulting in more stable training.
Reason: Your scene has complex trees and textures; frequent refinement often results in "stuttering clouds."

--warmup-length 800 (default 500, delays refine start)
Purpose: To stabilize the initial Gaussian before densifying/refining.
Reason: Refining too early can solidify initial errors into noise.

--reset-alpha-every 40 (default 30, reduces reset frequency)
Purpose: Prevents alpha from being broken down too frequently, reducing flickering instability.
Reason: The scene is already large; a slightly "gentler" approach usually results in a cleaner image.

--stop-screen-size-at 3000 (default 4000, stops large-scale splits earlier)
Purpose: Stops large-screen-size Gaussian splits earlier, suppressing large floating artifacts.
Reason: Blurred patches/overlays in your image often worsen around these types of Gaussian splits.

--ssim-weight 0.15 (default 0.2, slightly reduced)
Purpose: Reduces the "blurring/smearing" effect caused by overly strong structural similarity constraints.
Reason: In aerial and vegetated scenes, slightly reducing SSIM often results in cleaner edges.

--save-every 2000 + --val --val-render ... (Adds evaluation and intermediate results)
Purpose: Allows reviewing 2k interval snapshots and validation images, facilitating the selection of the optimal stopping point.
Reason: You've already seen that "more iterations aren't always better," requiring comparable intermediate checkpoints.

In short: The core idea behind this parameter set is **"less late-stage growth + slightly earlier convergence**," prioritizing the reduction of your most **noticeable floating noise and smearing effect.**

Explanation of adjusting parameter from AI

After understanding the principles of adjusting each parameter, I started to optimize the detailed artifacts. I gave the screenshot of the detailed problem I wanted to solve to AI, and used the parameters it provided to do several rounds of optimization to achieve the final, relatively high-precision Gaussian model.

Final outcome

Section 5: Learning Growth

At the beginning of the class, I could not even believe I could survive the whole teaching process. Not only because of the massive unfamiliar abstract knowledge, but also because I need to run some coding projects by myself using Docker and ROS, which I have never used before. But at the end of the semester, I know how to learn a new project from GitHub, pull the Docker image, and even debug the code to get better results. I think the most valuable thing is that this class just showed me a completely amazing way to learn a new project and technology with AI assistance, which will be really helpful to my future career development.

Section 6: Critical Reflection

I think AI is more like a partner or patient teacher during my learning process, rather than an employee who could do all the tasks for me. We debug, optimize, and finish the projects together, and the most useful part is that it can explain some problems in really easy-to-understand words, which really deepens my understanding of the whole computer visual process.

Section 7: Evidence

Shown in the section 4: Problem-Solving Process